1.0 GENERAL

1.1. WHEN ADOPTED IN THIS DOCUMENT OR IN OTHER CONTRACT DOCUMENTS, THE LATEST EDITION OF THE FOLLOWING CODES, STANDARDS, SPECIFICATIONS AND REFERENCES IN EFFECT ON THE DATE OF CONTRACT AWARD SHALL BE USED,EXCEPT AS OTHERWISE NOTED. SHORT TITLES WILL BE USED HEREIN WHEN APPROPRIATE.

a. MD-101-SMEM-EG-ST-SPC-0001 - STEELWORK FABRICATION
b. MD-101-SMEM-EG-ST-SPC-0002 - STEELWORK ERECTION
c. MD-101-SMEM-EG-ST-SPC-0003 - GROUT
d. MD-101-SMEM-EG-ST-SPC-0009 - PROTECTIVE COATINGS FOR STEELWORK

1.2. DESIGN DRAWINGS SHALL NOT BE SCALED. ANY DISCREPANCIES WITHIN THE DOCUMENTATION SHALL BE REFERRED TO THE EMPLOYER FOR CLARIFICATION BEFORE PROCEEDING.

1.3. PRODUCTS EQUIVALENT TO THOSE SPECIFIED ON THE DESIGN DRAWINGS MAY BE USED ONLY WITH THE APPROVAL OF THE EMPLOYER.

1.4. ALL DIMENSIONS ARE GIVEN IN MILLIMETRES.

1.5. ALL LEVELS ARE GIVEN IN METRES.

1.6. VERTICAL DATUM IS MEAN SEA LEVEL.

1.7. HORIZONTAL DATUM IS TO BE SPECIFIED.

1.8. ALL FABRICATED STEELWORK EDGES SHALL BE FREE FROM BURRS, SHARP CORNERS, FINS, GOUGES AND OTHER DEFECTS.

1.9 FOR EARTHING PROTECTION REQUIREMENTS OF STEELWORK STRUCTURES REFER TO ELECTRICAL DESIGN DRAWINGS AND ELECTRICAL STANDARD DETAILS ON DRAWING MD-101-SMEM-EG-EL-EST-0001.

1.10 PROJECT SPECIFIC NOTES REFER DESIGN DRAWING.

2.0 BOLTING

2.1. ALL BOLTS, NUTS AND WASHERS SHALL BE HOT DIPPED GALVANIZED TO SPECIFICATION MD-101-SMEM-EG-ST-SPC-0009.

2.2. ALL BOLTED CONNECTIONS FOR STRUCTURAL APPLICATIONS SHALL BE M20 HIGH STRENGTH AND COMPLY WITH NOTE 4.4.

2.3. ALL BOLTED CONNECTIONS ARE SNUG TIGHT (ST) UNO. FULLY TENSIONED BEARING (PT) AND SLIP-CRITICAL (SC) BOLTING SHALL BE SPECIFIED ON DESIGN DRAWING WHERE REQUIRED.

2.4. ALL STANDARD BOLTS SHOWN ON DESIGN DRAWINGS & STANDARD DETAILS SHALL COMPLY WITH NOTE 4.5.

2.5. WHERE BOLTS OTHER THAN M20 ARE SPECIFIED THE DETAILS SHOWN ON STANDARD DRAWINGS SHALL APPLY WITH ADJUSTMENTS FOR EDGE DISTANCES, SPACING, CLEARANCES ETC. MADE AS REQUIRED.

2.6. HOLES FOR BOLTS SPECIFIED ON STRUCTURAL AND MECHANICAL DRAWINGS SHALL BE 2 mm GREATER THAN THE BOLT DIAMETER, EXCEPT FOR BOLTS M24 OR GREATER WHERE HOLES SHALL BE 3mm GREATER THAN BOLT DIAMETER. HOLES IN BASE PLATES FOR ANCHOR BOLTS SHALL BE:
a. BOLT DIAMETER + 6mm FOR BOLTS UP TO AND INCLUDING M24.
b. BOLT DIAMETER + 8mm FOR BOLTS OVER M24

2.7. BOLT HOLES SHALL BE POSITIONED ON GAUGE LINES OF THE STEEL SECTIONS. BOLT HOLE SETOUT FOR ANGLES AND CHS MEMBERS REFER STANDARD DRAWING MD-101-SMEM-EG-ST-DST-0012

2.8. TIGHTENING OF BOLTS SHALL BE CARRIED OUT USING THE PART TURN METHOD. BOLTS THAT HAVE BEEN FULLY TENSIONED SHALL BE MARKED FOR VISUAL RECOGNITION.

2.9. REFER TO MECHANICAL AND VENDOR DRAWINGS FOR EQUIPMENT BOLTING DETAILS.

2.10. ALL BOLT HOLES SHALL BE PUNCHED OR DRILLED. HOLE PUNCHING AND SLOTTED HOLES SHALL BE IN ACCORDANCE WITH MD-101-SMEM-EG-ST-SPC-0001.

2.11. PROPRIETARY ANCHORS SHALL BE M20 CHEMICAL ANCHORS, INSTALLED STRICTLY IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS AND COMPLY WITH ANCHOR BOLT SPECIFICATION - MD-101-SMEM-EG-CV-SPC-0002.

3.0 GROUTING

3.1. ALL COLUMN BASEPLATES AND STEEL TO CONCRETE INTERFACES SHALL BE GROUTED IN ACCORDANCE WITH THE STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-003.

3.2. THE NOMINAL GROUT THICKNESS UNDER COLUMN BASEPLATES SHALL BE 30 mm. GROUT THICKNESS FOR EQUIPMENT SHALL BE AS SPECIFIED BY EQUIPMENT MANUFACTURER. GROUT SHALL BE CURED IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.

3.3. STEEL PACKERS UNDER BASEPLATES SHALL BE FULLY ENCASED WITH GROUT. GROUT COVER TO PACKERS AT BASE PLATE EDGES SHALL BE 50mm MINIMUM.

3.4. BLOCK OUTS, SHEAR KEY RECESSES & SLEEVES FOR ANCHOR BOLTS SHALL BE FILLED WITH GROUT.

4.0 MATERIALS

4.1. STRUCTURAL HOT ROLLED AND WELDED STEEL SECTIONS SHALL BE ASTM A36 AND COMPLY WITH AISC LRFD STEEL CODES.

4.2. STRUCTURAL PIPE SECTIONS SHALL BE ASTM A53 TYPE E OR S, GRADE B OR ASTM A106, GRADE B.

4.3. SQUARE HOLLOW SECTIONS SHALL BE ASTM A500 GRADE B.

4.4 HIGH STRENGTH BOLTS FOR STRUCTURAL APPLICATIONS SHALL BE ASTM A325/A325M TYPE 1 OR ASTM A490/A490M, UNLESS SPECIFIED OTHERWISE ON DESIGN DRAWING. WASHERS TO ASTM F436/F436M. DIRECT TENSION INDICATOR (DTI) WASHERS TO ASTM F959/F959M WHEN SPECIFIED ON DESIGN DRAWINGS HEAVY HEX NUTS TO ASTM A563/A563M GRADE DH.

4.5 STANDARD BOLTS FOR NON STRUCTURAL APPLICATIONS SHALL BE ASTM A307 GRADE A HEAVY HEX. WASHERS SHALL BE ASTM F436/F436M. HEAVY HEX NUTS SHALL BE ASTM/A563M GRADE A.

5.0 WELDING

5.1. ALL WELDS AND WELDING PROCEDURES SHALL BE IN ACCORDANCE WITH THE PROJECT SPECIFICATION MD-101-SMEM-EG-ST-SPC-001 AND AWS D1.1 AND FABRICATOR SHALL BE FULLY APPROVED BY AWS D1.1.

5.2. WELDING CONSUMABLES SHALL HAVE:
a. AN ELECTRODE STRENGTH OF 400MPa MINIMUM YIELD STRESS AND 480MPa MINIMUM TENSILE STRENGTH.
b. E60XX ELECTRODES MAY BE USED FOR TACK WELDING.

5.3. ALL WELDS SHALL BE FILLET WELDS AND SHALL BE:
6mm CONTINUOUS FOR 12PL AND LESS
8mm CONTINUOUS FOR 16PL AND 20PL
10mm CONTINUOUS FOR 25PL AND OVER
(WELDS APPLY TO THE THINNEST PLATE IN THE CONNECTION)

5.4. ALL WELDS TO VERTICAL BRACING CLEATS SHALL BE:
6mm CONTINUOUS FOR 10PL AND LESS
8mm CONTINUOUS FOR 12PL
10mm CONTINUOUS FOR 16PL AND 20PL
12mm CONTINUOUS FOR 25PL AND OVER

5.5. ALL BUTT WELDS SHALL BE FULL PENETRATION AND SHALL BE CONTINUOUS.

5.6. INSPECTION AND PERMISSIBLE LEVEL OF DEFECT SHALL BE IN ACCORDANCE WITH THE FOLLOWING:
a. 100% VISUAL INSPECTION TO DETERMINE IF WELDS REQUIRED ON DRAWINGS HAVE BEEN INCLUDED AND COMPLY WITH ACCEPTANCE CRITERIA AS SET OUT IN TABLE 6.1, AWS D1.1.
b. 5% OF GROOVE, BUTT AND FILLET WELDS SHALL BE TESTED WITH MAGNETIC PARTICLE, LIQUID PENETRATOR RADIOGRAPHY AS PER AWS D1.1.
DEFECT ACCEPTANCE CRITERIA SHALL BE IN ACCORDANCE WITH SECTION 6, AWS D1.1.
c. FOR PLATES WITH THICKNESS OF NOT LESS THAN 20mm, 5% OF EACH MULTIPASS BUTT WELD SHALL BE EXAMINED ULTRASONICALLY AS PER AWSD1.1 TO DETERMINE WHETHER THERE ARE ANY TRANSVERSE CRACKS WITHIN THE WELD METAL.
d. SUBJECT TO APPROVAL BY THE EMPLOYER, SUCCESSFUL COMPLIANCE WITH EACH TEST SHALL ALLOW A PROGRESSIVE REDUCTION IN TESTING FREQUENCY.

5.7. UNLESS SHOWN AS A BOLTED CONNECTION, ALL PLATES AND SECTIONS INDICATED IN CONTACT SHALL BE WELDED ALL ROUND.

5.8. WHERE PAINTING IS SPECIFIED AS THE CORROSION PROTECTION SYSTEM, ALL TUBULAR SECTIONS SHALL BE FULLY SEALED. SEAL PLATES SHALL BE 5 mm THICK, SEALED ALL ROUND.

6.0 CONNECTIONS AND SPLICES

6.1. STRUCTURAL CAPACITIES OF THE CONNECTIONS SHOWN ON THESE DRAWINGS SHALL BE CHECKED BY THE CONTRACTOR TO DETERMINE THEIR SUITABILITY FOR THE INTENDED USE.

6.2. ALL BOLTED CONNECTIONS SHALL HAVE A MINIMUM OF 2 BOLTS PER CONNECTION.

6.3. SPLICES IN MEMBERS SHALL BE PROVIDED ONLY WHERE SHOWN ON THE DESIGN DRAWINGS. SHOULD ADDITIONAL SPLICES BE REQUIRED THE CONTRACTOR SHALL APPLY FOR APPROVAL FOR EACH SPLICE NOT SHOWN ON THE DRAWINGS. SPLICES SHALL NOT BE PROVIDED WHERE CRITICAL BENDING STRESSES ARE DEVELOPED.

6.4. ALL BEAM TO BEAM CONNECTIONS SHALL BE:
FULL END PLATE FOR ALL MEMBERS WITH FLOOR BRACING CONNECTED TO THE END OF A BEAM.
ANGLE CLEAT CONN. FOR ALL STAIR STRINGERS.
WEB SIDE PLATE FOR ALL OTHER MEMBERS.

6.5. ALL BEAM TO COLUMN CONNECTIONS SHALL BE FULL END PLATES.
FOR END PLATE EXTENSION & PLATE THICKNESS REQUIREMENTS WHERE VERTICAL BRACING OCCURS REFER MD-101-SMEM-EGST-DST-0013 'TYPICAL BRACE CLEAT SETOUT'.

6.6. CONNECTIONS SHALL BE CONCENTRIC USING CENTRE OF AREA, EXCEPT FOR:
CHANNELS-USE BACK OF CHANNEL.
BRACING-USE CENTRE LINE OF LEG OR GAUGE LINE.

6.7. ALL CLEATS, GUSSETS, END AND STIFFENER PLATES SHALL BE MINIMUM 10mm THICK UNLESS NOTED OTHERWISE.

6.8. ALL COLUMNS SHALL BE IN FULL BEARING CONTACT WITH BASEPLATES. THE ENDS OF THE COLUMN SHALL BE COLD SAWN OR MACHINE CUT IN ACCORDANCE WITH SAUDI BUILDING CODE SBC 306 SECTION 13 TO ACHIEVE 'FULL BEARING CAPACITY'.

6.9 BASEPLATE CONNECTIONS TO COLUMNS:

TO 'W' SHAPE & CHANNEL COLUMNS:

COLUMNS 200mm DEEP AND LARGER - BUTT WELDS ON FLANGES AND CONTINUOUS FILLET WELD ON BOTH SIDES OF WEB REFER NOTE '5.0 WELDING'.
COLUMNS LESS THAN 200mm DEEP - CONTINUOUS FILLET WELDS ALL AROUND REFER NOTE '5.0 WELDING'.

TO ANGLE COLUMNS:

COLUMNS 200mm DEEP - BUTT WELD.
COLUMNS LESS THAN 200mm DEEP - 6mm CONTINUOUS FILLET WELDS ALL AROUND - REFER NOTE '5.0 WELDING'.

TO PIPE & SQUARE HSS COLUMNS:

HOLLOW SECTIONS WITH WALL THICKNESS LESS THAN OR EQUAL TO 5.0mm SHALL HAVE 6mm FILLET WELD ALL AROUND.
HOLLOW SECTIONS WITH WALL THICKNESS GREATER THAN 5.0mm SHALL HAVE BUTT WELDS.

7.0 MONORAILS

7.1. MONORAILS SHALL BE CONSTRUCTED IN ACCORDANCE WITH ASME B30.11 AND STATUTORY REGULATIONS.

7.2. PACKING TO BE LAMINATED MILD STEEL PLATES. LAMINATED PACKS TO BE WELDED TOGETHER ON SITE.

7.3. MONORAILS SHALL BE FINISHED PAINTED PURE ORANGE RAL 2004.

7.4. PAINT SIGN ON BOTH SIDES OF MONORAIL WEB INDICATING WORKING LOAD LIMIT - (eg 2 TONNE WLL) IN BLACK LETTERS 100 mm MINIMUM HIGH, IN A POSITION LEGIBLE FROM THE OPERATING PLATFORM AND OTHER INFORMATION AS MAY BE REQUIRED BY STATUTORY AUTHORITY.

7.5. MONORAILS SHALL BE INSTALLED LEVEL TO WITHIN A TOLERANCE OF +/- 3mm.

8.0 PLATFORMS, WALKWAYS, STAIRWAYS AND LADDERS

8.1. ALL FIXED PLATFORMS, WALKWAYS, STAIRWAYS AND LADDERS SHALL BE IN ACCORDANCE WITH THESE STANDARD DRAWINGS AND MD-101-SMEM-EGST-DST-0002-11, 12, 13, 14 & 15

8.2. ALL FLOOR GRATING SHALL BE MINIMUM WEBFORGE A255 OR SIMILAR APPROVED.

8.3. ALL EDGES OF FLOOR GRATING INCLUDING CUT-OUTS SHALL BE FULLY BANDED.

8.4. ALL STAIR TREADS SHALL BE "WEBFORGE" TYPE T6/A255MPG.

8.5. WHERE FLOOR SLOPE IS GREATER THAN OR EQUAL TO 7 DEGREES ALL FLOOR GRATING SHALL HAVE 10 mm SQUARE OR SIMILAR APPROVED MILD STEEL BARS WELDED AT RIGHT ANGLES TO THE SLOPE AND AT 300 CENTRES ON THE TOP SURFACE.

8.6. GRATING INSTALLATION TOLERANCES SHALL BE TO MANUFACTURER'S RECOMMENDED DETAILS.

8.7. REMOVABLE GRATING PANELS SHALL WEIGH 20kg MAXIMUM AND SHALL BE FIXED DOWN WITH A MINIMUM OF 4 FASTENINGS FIXED TO MANUFACTURER'S RECOMMENDATIONS.

8.8. PRIOR TO CUTTING OF ANY EXISTING GRATING THE CONTRACTOR MUST SEEK APPROVAL FOR CUTTING AND ALSO SUBMIT A PROCEDURE FOR REINSTATEMENT OF GRATING BEFORE ANY WORK MAY COMMENCE.

8.9. ALL OPENINGS IN GRATED AND PLATED FLOORS, UNGUARDED BY HANDRAIL SHALL HAVE KICKPLATE INSTALLED.

8.10. ALL FLOOR PLATE SHALL HAVE A RAISED ANGULAR PATTERN, BE 6mm THICK AND BE TO ASTM A786/A786M WITH A GALVANIZED FINISH (REFER NOTE '9.0 SURFACE PROTECTION OF STEEL').

8.11. EXTENT OF FLOOR GRATING DENOTED

8.12. EXTENT OF FLOOR PLATE DENOTED

8.13. DIRECTION OF FLOOR GRATING LOAD BARS DENOTED

8.14. ALL STAIRS SHALL HAVE BOLTED STAIR TREADS. BOLTS SHALL BE M12 STANDARD BOLTS. (WHERE STAIRS CONNECT INTO STRUCTURES WHICH ADOPT HIGH STRENGTH (PT) BOLTED CONNECTIONS PROVIDE LOCK NUTS TO BOLTS OR PUNCH THREADS OF BOLTS IN TWO PLACES AT EACH BOLT TO PREVENT NUT WORKING LOOSE).

8.15. ALL HANDRAILS SHALL BE DESIGN APPROVED TUBULAR TYPE, HOT DIPPED GALVANIZED AND INSTALLED AS PER MANUFACTURERS RECOMMENDATIONS.

8.16. ALL HANDRAIL STANCHIONS SHALL BE:
a) TOP MOUNT - FIXING TO STEEL STRUCTURE.
b) ANGLE MOUNT - FIXING TO ANGLED STAIR STRINGERS.

8.17. PROVIDE DESIGN APPROVED SELF CLOSING GATES OR SIMILAR APPROVED FIXED TO HANDRAILS AT THE TOP OF ALL LADDERS.

9.0 SURFACE PROTECTION OF STEEL

9.1. WHERE PAINTING IS SPECIFIED AS THE CORROSION PROTECTION SYSTEM, THE SURFACE PREPARATION AND PROTECTION SHALL BE IN ACCORDANCE WITH THE STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0009.

9.2. WHERE GALVANIZING IS SPECIFIED AS THE CORROSION PROTECTION SYSTEM IT SHALL BE IN ACCORDANCE WITH THE STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0009.

9.3. WHERE GALVANIZING IS SPECIFIED WITH AN APPLIED TOP COAT IT SHALL BE IN ACCORDANCE WITH THE STANDARD SPECIFICATION MD-101-SMEM-EG-ST-SPC-0009.

10.0 SHOP DRAWINGS

10.1. SHOP DRAWINGS SHALL BE SUBMITTED TO THE EMPLOYER FOR APPROVAL PRIOR TO FABRICATION OF STRUCTURAL STEEL.

10.2. APPROVAL OF SHOP DRAWINGS DOES NOT RELIEVE THE FABRICATOR OF HIS RESPONSIBILITY FOR DETAIL DIMENSIONS ON SHOP AND ERECTION DRAWINGS, THE GENERAL FIT-UP ON PARTS TO BE ASSEMBLED IN THE FIELD, NOR FOR COMPLIANCE WITH DESIGN DRAWINGS.

10.3. CHANGES IN DETAILS OR SUBSTITUTION OF MEMBERS SHALL NOT BE MADE WITHOUT PRIOR APPROVAL OF THE EMPLOYER.

10.4. WHERE STEELWORK IS GALVANISED THE DETAILING OF STEELWORK SHALL PROVIDE FOR ALL DRAIN AND VENT HOLES AS REQUIRED.

11.0 ERECTION

11.1. ERECTION OF STRUCTURAL STEELWORK SHALL BE IN ACCORDANCE WITH THE DESIGN DRAWINGS, AND STANDARD SPECIFICATION-MD-101-SMEM-EG-ST-SPC-0002.

11.2. PRIOR TO ERECTION, THE CONTRACTOR SHALL PROVIDE A CONSTRUCTION METHODOLOGY INCLUDING, BUT NOT LIMITED TO, DETAILS OF ANY TEMPORARY PROPS AND BRACES. THIS CONSTRUCTION METHODOLOGY, INCLUDING ALL SUPPORTING DETAILS, SHALL BE SUBMITTED IN WRITING TO THE EMPLOYER FOR AGREEMENT AND APPROVAL BEFORE COMMENCEMENT OF SITE WORKS.

11.3. DURING STORAGE AND ERECTION THE CONTRACTOR SHALL ENSURE THAT STRUCTURAL MEMBERS ARE NOT OVERSTRESSED, DAMAGED OR DEFORMED.

11.4. WHERE NOMINATED ON THE DESIGN DRAWINGS, PACKING SHALL BE LAMINATED MILD STEEL PLATES THAT SHALL BE WELDED TOGETHER FOLLOWING THE TENSIONING OF THE BOLTS.

11.5. THE CONTRACTOR MAY PRE-ASSEMBLE STRUCTURAL MODULES AND PLATFORMS, COMPLETE WITH HANDRAILS AND GRATING SUBJECT TO THE APPROVAL OF LIFTING AND ERECTION PROCEDURE SUBMITTED TO THE EMPLOYER.

11.6. STATIC LINE ANCHORAGE POINTS FOR FALL ARRESTING DEVICES SHALL BE PROVIDED ON ALL STRUCTURAL STEELWORK.
FOR GENERAL STRUCTURES PROVIDE 22 DIAMETER HOLES 2.1m ABOVE THE FLOOR OR SUPPORT LEVEL:
-EXTERNAL COLUMNS - INSIDE FLANGE (ANY SIDE OF WEB)
-INTERNAL COLUMNS - BOTH FLANGES (ANY SIDE OF WEB)
50mm BELOW THE TOP OF COLUMN.
STATIC LINES, ANCHORAGE AND ASSOCIATED SAFETY EQUIPMENT SHALL BE DESIGNED AND INSTALLED BY THE ERECTION CONTRACTOR FOR EACH LOCATION.

12.0 PURLINS AND GIRTS

12.1 COLD FORMED STEEL PURLIN & GIRT SECTIONS SHALL BE ASTM 1008 GRADE 50.

12.2 PURLINS & GIRTS SHALL BE FIXED WITH HIGH STRENGTH BOLTS.

12.3 FASTENERS, FLASHING AND METHODS USED FOR FIXING CLADDING TO PURLINS & GIRTS SHALL BE IN ACCORDANCE WITH THE SHEETING MANUFACTURER'S RECOMMENDATIONS.

13.0 FIRE PROOFING

13.1 FIRE PROOFING PROCEDURE AND MATERIAL SHALL COMPLY WITH SBC 801

THE STRUCTURAL STEELWORK STANDARD NOTES ON THIS DRAWING APPLY UNLESS NOTED OTHERWISE ON THESE STANDARD DRAWINGS OR ON THE DESIGN DRAWINGS

NOTES:

1. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH CONCRETE STANDARD NOTES ON DRAWING MD-101-SMEM-EG-ST-DST-0001.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD

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MA'ADEN معادن Saudi Arabian Mining Company

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MA'ADEN		
SCALE NTS		A1

MA'ADEN ENGINEERING MANUAL
ALL AREAS
STRUCTURAL STANDARDS
STEELWORK GENERAL NOTES

DRAWING NUMBER MD-101-SMEM-EG-ST-DST-0009 REV C01

MA'ADEN PHOSPHATE PROJECT-A1 BORDER-REV A02
$DATE$ $TIME$ $FILE$